
Capital Flight and External Debt in Nigeria

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Over the last few years, many heavily indebted countries have experienced heavy capital outflows or capital flight, and capital flight has become an important topic for detailed study in some of these countries. This interest arises, among other reasons, because of the role that external assets stored away in foreign lands could play if left in the domestic economy, to augment the dwindling resources from international creditors.

It is widely believed that the study of capital flight from highly indebted countries is important because of the economic problems such flight can create (see Khan, 1989). The outflow of capital can cause a shortage of liquidity in the economy and lead to upward pressure on the interest rate. Similarly, the shortage of liquidity can cause a depreciation of the domestic currency if the authorities are operating a floating exchange rate system. If the government is defending a particular exchange rate, a loss of reserves will ensue.

The loss of resources has several long-term effects. The first is that the availability of resources for domestic investment is reduced. The rate of capital formation is reduced by capital flight, and this adversely affects the country's current and future prospects. Income that is generated abroad as well as wealth held abroad are outside the purview of relevant authorities and cannot be taxed. The effects are a reduction in government revenue and government debt-servicing capacity. Capital flight can exacerbate a balance of payments crisis if, at the time the balance of payments difficulty is being experienced, capital outflows are taking place. Capital flight can also compound the foreign finance problems of heavily indebted countries if creditors are reluctant to give further assistance as a result of capital outflows.

The link between capital flight and growth is expressed vividly in the literature. Two of the relevant works are Deppler and Williamson (1987) and Lessard and Williamson (1987). The linkage is expressed as follows by Deppler and Williamson (1987, p. 52):

the fundamental economic concern about capital flight, however, is that it reduces welfare in the sense that it leads to a net loss in the total real resources available to an economy for investment and growth. That is, capital flight is viewed as a diversion of domestic savings away from financing domestic real

investment and in favor of foreign financial investment. As a result, the pace of growth and development of the economy is retarded from what it otherwise would have been.

The linkage is expressed by Lessard and Williamson (1987, p. 224) in the same vein:

The best case involves a reduction in the savings to finance domestic investment of a magnitude essentially equal to the size of the capital flight. Future growth will in consequence be lower. The worst case involves a reduction not just in future growth possibilities but also in the current level of output by some multiple of the size of capital flights.

A brief critique of the views expressed is given later.

Many reasons are often adduced for capital flight. The preponderant causes are economic, but the economic aspects are inextricably interwoven with political causes and favorable foreign economic incentives. Domestic macroeconomic policy distortions are a main factor. These distortions manifest themselves in large public sector deficits, exchange rate misalignment, and financial repression. Apart from these, there are also the incentives provided by foreign banks and governments. Part of the explanation for capital flight is also political. This is predicated on corruption and access to foreign funds by political leaders. It has been alleged that some political leaders, through the perquisites of their offices, siphon funds to foreign countries.¹

Nigeria is one of the heavily indebted countries where the issue of capital flight has been regarded as important. There is, however, no comprehensive study on the causes, measurement, magnitude, and consequences of capital flight in Nigeria.² Given the magnitude of Nigeria's external debt and the possible impact of capital flight on the country's real debt-service capacity, a study of capital flight and external debt is essential.

This chapter intends to take up that challenge.³ It focuses on the definition, magnitude, determinants, mechanisms, and consequences of capital flight. The analysis also explores possible measurements of assets in which the money is held once it arrives abroad. The emphasis

¹For the developing countries, examples often cited included the late Ferdinand Marcos, president of the Philippines, and former leaders of Haiti and Zaïre (now the Democratic Republic of the Congo). In Nigeria, some powerful politicians who fled the country after the military coup of 1983 are alleged to belong to the same category.

²The available studies that refer to Nigeria are limited. The comprehensive study by Rojas-Suárez (1991) lumped Nigeria in the group of heavily indebted countries. The study did not estimate capital flight for each country.

³A survey of the various methods undertaken by Deppler and Williamson (1987) lists only four of the methods discussed here.

is on the macroeconomic effects of capital flight within the context of economic, socioeconomic, and other functions.

Specifically, the chapter

- examines the magnitude of capital flight for the period 1970–88 and a number of alternative methods of its measurement;
- analyzes the economic (mainly macroeconomic) and other factors responsible for capital flight;
- examines the conduits through which capital flight takes place
- identifies the major consequences of capital flight for the domestic economy; and
- examines the linkages between capital flight and external debt and draws policy conclusions.

The Definition of Capital Flight

This section surveys and analyzes the various definitions and measures of capital flight in the existing literature. The approach adopted is twofold. First is a discussion at the conceptual level, exploring the rationale or basis for classifying domestic outflows as capital flight instead of normal flows. The second approach is strictly empirical. The objective is to compute and analyze alternative measures (estimates) of capital flight. The measurements are derived from a common database for the period 1970–88 to show the variation in the estimates under alternative definitions.

The use of the term *capital flight* arouses strong emotions. Some analysts view capital flight as a symptom of a sick society. Some observers see it as the cause of the heavily indebted countries' inability to recover from their problems. Capital flight is regarded by others as an unnecessarily pejorative description of natural, economically rational responses to the portfolio choices that have confronted wealthy residents of some debtor countries in recent years (Lessard and Williamson, 1987, p. 201). The controversy surrounding the term is due partly to the absence of a precise and universally accepted definition for it, and partly due to the way the term is used between developed and developing countries. It is usual among some economists to refer to capital outflows from developed countries as foreign investment, whereas the same activity undertaken by the residents of a developing country is referred to as capital flight.

One of the reasons for this dichotomy is the belief that investors from the developed economies are responding to better opportunities abroad. The investors from developing countries, on the other hand,

are said to be escaping the high risks they perceive at home. This interpretation makes it obvious why a lot of economists are ill at ease with this definition of capital flight. In general, it is believed that investors from all countries, whether developed or developing, will base their decisions on the relative returns and risks of investments at home and abroad.

A distinction is also often made between legal and illegal transactions in distinguishing between capital flight and so-called normal capital outflows. Since illegal transactions are not reported to the compilers of the balance of payments statistics, it is difficult to know the extent to which they therefore constitute capital flight. Capital flight is capital that flees (Walter, 1986; Kindleberger, 1987). Alternatively, capital outflows in response to economic or political crises are capital flight (Husted and Melvin, 1990). Normal capital flows, on the other hand, refer to flows that correspond to ordinary portfolio diversification on the part of domestic residents.

According to Cuddington (1986, p. 2), capital flight refers to short-term capital outflows. It involves hot money that responds to political or financial crises, heavier taxes, a prospective tightening of capital controls, a major devaluation of the domestic currency, or actual or incipient hyperinflation. Morgan Guaranty Trust Company (1986, p. 13) defines capital flight as the reported or unreported acquisition of foreign assets by the nonbank private sector and elements of the public sector.

To classify our thoughts on capital flows, Table 1 presents a taxonomy of factors explaining international capital flows used by Lessard and Williamson (1987). The upper left section of the table identifies various factors based on differences in economic returns across countries. In the upper right section are those additional factors that deal with two-way flows, or normal portfolio diversification. Most theoretical and empirical studies of capital flight have emphasized the lower left and right sections. The factors emphasized are those that create a wedge between economic and financial returns, regardless of whether they operate across the board or asymmetrically between residents and nonresidents (Lessard and Williamson, 1987).

From Table 1 it can be seen that normal capital outflows are those that seek to maximize economic returns and opportunities between countries. Normal portfolio diversification takes place on the basis of differentials in economic returns. Capital flight, on the other hand, as seen from this analysis, is that subset of capital outflows that is propelled by source country policies (Lessard and Williamson, 1987, p. 217).

Table 1. Taxonomy of Factors Explaining International Capital Flows

	One-Way Flows	Two-Way Flows
Economic risks and returns	Natural resource endowments Terms of trade Technological changes Demographic shifts General economic management	Differences in absolute riskiness of economies Low correlation of risky outcomes across countries Differences in investor risk preferences
Financial risks and returns	Taxes (deviations from world levels) Inflation Default on government obligations Devaluation Financial repression Taxes on financial intermediation Political instability, potential confiscation	Differences in taxes and their incidence between residents and nonresidents Differences in nature and incidence of country risk Asymmetric application of guarantees Different interest ceilings for residents and nonresidents Different access to foreign exchange-denominated claims

Source: Lessard and Williamson (1987, p. 216).

A Review of the Measures and Estimates of Capital Flight

A number of estimates of capital flight have been made over the last several years. The preponderance of these studies focuses on Argentina, Brazil, Chile, Korea, Mexico, Peru, the Philippines, and Venezuela. A study by Rojas-Suárez (1991) covers Argentina, Bolivia, Chile, Colombia, Ecuador, Gabon, Jamaica, Mexico, Nigeria, Peru, the Philippines, Venezuela, and the former Yugoslavia. These studies differ from one another in terms of methodology, country coverage, data sources, and time span. The most significant of the studies on capital flight that have made an impact are those by Dooley (1986), Dooley and others (1986), World Bank (1985), Morgan Guaranty Trust Company (1986), Cuddington (1986), Cumby and Levich (1987), Gulati (1985), Lessard and Williamson (1987), Khan and Ul Haque (1985), and Khan (1989).

In the World Bank (1985) study, capital flight is defined as the sum of gross capital outflows and the current account deficit less increases in official foreign reserves. Cuddington (1986) takes a different approach. Capital flight is defined as short-term speculative outflows; according to him, this is the typical meaning of capital flight,

and the term therefore refers to short-term external assets held by the nonbank private sector plus errors and omissions in the balance of payments. Cuddington's (1986) approach is concentrated on so-called hot money funds, because these funds respond quickly to changes in expected returns or to changes in risk. Variations in economic conditions are likely to affect the flow of these funds, which are expected to return very quickly to the home country when economic conditions improve.

Khan and Ul Haque (1985) calculated capital flight for eight highly indebted developing countries for the period 1974–82. Capital flight is defined in two ways. First, it is defined simply as gross private short-term capital flows plus net errors and omissions in the country's balance of payments accounts. This is the same as Cuddington's definition. The second method tries to take account of normal capital flows. Capital flight is defined as that part of the increase in external claims that yields no recorded investment income (see Dooley, 1986).

The Dooley (1986) study does attempt to tease out the normality of capital flows by specifically separating normal and abnormal capital flows. The Dooley method seeks to measure the stock of privately held foreign assets that do not generate income reported to the domestic authorities. The identified capital outflows in the balance of payments accounts are cumulated, and three adjustments are then made to capture the unreported capital flows. First, errors and omissions are added. Second, a comparison is made between the stock of external debt as reported in the World Bank data and those reported *in the balance of payments statistics*. The difference between the two is added to the estimate of the increase in private sector foreign assets. The third adjustment involves the calculation of the stock of external assets needed to give the observed amount of investment income in the balance of payments at international market rates, for example the U.S. treasury bill rate.

In the Morgan Guaranty Trust Company (1986, p. 13) study, capital flight is defined as the reported or unreported acquisition of foreign assets by the nonbank private sector and some elements of the public sector. Capital flight is therefore net investment inflows plus changes in gross external debt plus the current account balance and changes in selected gross foreign assets. Cline (1986) critiques this definition. He argues that income from tourism and border transactions should be subtracted, since these earnings are beyond the control of the relevant foreign exchange authorities. He also argues that reinvested investment income should not be considered as capital flight, since this also is beyond the control of the authorities.

Table 2. Alternative Measures of Capital Flight

Definition	Methodology	Authors
Narrow measure	Net short-term capital outflows plus errors and omissions	Cuddington (1986)
Derived measure ¹	Part of the increase in external claims that yields no recorded investment income	Dooley (1986)
Residual measure or sources-and-uses approach	Change in debt plus net foreign and direct investment minus current account deficit plus change in reserves	Chang and Cumby (1991), World Bank (1985), Pastor (1990)
Private claims measure	Acquisition of external claims by the private sector, including deposit banks and the nonbank sector, plus recorded errors and omissions in the balance of payments	Conesa (1987)
Mirror stock statistics method	Cross-border bank deposits by residence of depositor	Khan and Ul Haque (1985)
Change in private foreign assets ²	The counterpart of the sum of net direct investment inflows, the change in gross external debt, the current account balance, and the change in selected gross foreign assets	Morgan Guaranty Trust Company (1986)

¹Also called the stock of unreported foreign assets measure.

²Also often seen as another aspect of the residual measure.

In what are referred to as "mirror stock statistics," capital flight is measured as the recorded cross-border bank deposits of nonbanks by residence of depositor. These figures can be found in the statistics published by the IMF.

Apart from these various measures, it is also possible to take cognizance of the misinvoicing of exports and imports in international trade. The derived results from trade misinvoicing can be added to any of the measures to derive another set of data on capital flight.

A schematic summary of the different definitions (or types of measure), the methodology, and the authors is presented for easy reference in Table 2. It should be emphasized that the different methods will yield different results, not only because the definitions are different but also because of differences in data sources. The results may therefore conflict, and comparisons are difficult to make.

Alternative Measures of Capital Flight for Nigeria

There are a number of objectives behind the calculation of the various alternative measures of capital flight for Nigeria. The primary objective, which is modest, is to show the range of capital flight estimates implied by the alternative definitions of capital flight that are offered. The additional objectives are to examine the extent to which capital flight is continuous or episodic. In other words, do heavy outflows tend to concentrate in certain years, and small flows in others? That finding in itself may suggest that capital flight is associated with particular economic or political events. In this realm, capital flight may be associated with poor macroeconomic management, large inflows of foreign exchange from exports (an export boom), changes between civilian and military governance, and so on.

As mentioned earlier, there is no precise and universally acceptable way of measuring capital flight. Some judgment is required, taking due cognizance of the objectives for measuring capital flight and the economic and social environment of the country for which capital flight is being measured. In this chapter a number of approaches are taken. The first recognizes that capital flight is speculative capital: it is hot money on the wing and can be expected to respond to the various forms of distortion mentioned earlier. Taking this approach, of course, means that capital flight refers essentially to capital exports by the private nonbank sector, although in some cases banks and official entities may also engage in it (Cuddington, 1986). Since capital flight is essentially concealed, it shows up in the errors and omissions of the balance of payments entry. Thus, capital flight is defined as the sum of short-term private capital flows plus errors and omissions in the balance of payments statistics. The results of the calculations are shown in Table 3.

As can be seen, some of the figures are negative. The intuitive explanation is that the negative signs cannot connote capital flight. As explained in the Cuddington study (1986, p. 5), the figures reflect capital flight net of unrecorded capital inflows.

Essentially, however, the years with negative signs are better perceived as years of capital repatriation, capital flight reversal, or capital inflows. Thus, in the period 1972–89, Nigeria's total capital inflow was \$7.6 billion. Out of this total, \$7.4 billion (97 percent) came in between 1972 and 1978, the period of Nigeria's oil boom. During the political era, that is, the period in which a civilian government was in power, the capital inflow was \$270 million. Thus, using this technique of calculation, one can see the episodic nature of capital flight. As a result of

Table 3. Nigeria: Capital Flight, 1970-89
(In millions of U.S. dollars)

	Capital Flight
1970	134.0
1971	205.0
1972	119.0
1973	-177.0
1974	48.0
1975	-42.0
1976	5.0
1977	-231.0
1978	43.0
1979	211.0
1980	-673.0
1981	106.0
1982	149.0
1983	-63.0
1984	-642.0
1985	-2,014.0
1986	-249.0
1987	-953.0
1988	-1,315.0
1989	-1,895.0
Totals	
1972-78	-7,362.0
1972-89	-7,573.0
1979-83	-270.0

Source: Author's calculations from data in International Monetary Fund (1990).

the buoyant economy associated with the oil boom years, the macro-economic environment can be said to be favorable to capital inflows.

Using approaches by Cumby and Levich (1987), Varman (1989), and Varman-Schneider (1991), we calculate from the balance of payments statistics a number of capital flight estimates for Nigeria using various methods shown in Table 4. The data are drawn mainly from the IMF's *International Financial Statistics Yearbook* and *Balance of Payments Statistics Yearbook* and from the World Bank's *World Debt Tables*. The result of the calculation is shown in Table 5. This is the first time that this has been done for Nigeria; there has been no similar calculation elsewhere using the different definitions shown in Table 4.

The differences in the magnitudes of the results using various definitions of capital flight are not surprising given the differences in definitions. The similarities and differences can be classified according to different periods. In 1970-73, for example, the World Bank (1985)

Table 4. Notations of Estimating Capital Flight

Notations	
A.	Current account balance
B.	Net foreign direct investment
C.	Private short-term capital outflows
D.	Portfolio investment
E.	Foreign assets of the banking system
F.	Changes in reserves
G.	Errors and omissions
H.	Changes in debt
I.	IMF credit
J.	Travel credit
K.	Reinvested income from foreign direct investment
L.	Other investment income
M.	Counterpart items

Capital Flight Estimates	
World Bank	$= (H + B + A + F)$
Erbe	$= (H + B + A + F)$
Morgan	$= (H + B + A + E + F)$
Cline	$= (H + B + A + E) - (J + K + L)$
Duwendag	$= (H + B + A + F + G + I + M)$

Source: Lessard and Williamson (1987), p. 38.

and Erbe (1985) estimates were generally lower than the other three estimates. The amplitudes of capital flight for the period 1985–87, however, were not too different in the four measures. What is important is the cumulative sums of capital flight for given specific years.

Of the four approaches, the Duwendag (1987) approach consistently gave the lowest estimates. A line graph of the approaches reveals that the Erbe, World Bank, and Duwendag approaches are close, as shown by the variations of these figures, just as the Morgan Guaranty and Cline estimates are very close in terms of the amplitude of the figures.

It is clear that the different results obtained derive from the choice of elements that go into the calculation of capital flight. It is also clear that all the approaches yield significant amounts of capital flight over the period covered.

Table 5. Nigeria: Alternative Estimates of Capital Flight, 1972–89
(In millions of U.S. dollars)

	Erbe (1985) and World Bank (1985)	Morgan Guaranty (1986)	Cline (1986)	Duwendag (1987)
1972	106.40	477.28	453.37	127.70
1973	636.10	1,265.38	1,228.03	551.75
1974	325.00	5,995.00	5,824.27	450.88
1975	119.80	5,988.60	5,474.48	148.04
1976	124.80	5,524.44	5,044.21	187.40
1977	2,490.00	7,021.86	6,554.79	2,111.95
1978	508.40	2,695.20	2,309.07	235.23
1979	-86.30	5,659.54	5,370.07	601.59
1980	2,713.30	12,974.11	12,234.36	2,590.79
1981	2,132.30	6,145.22	5,267.31	1,345.14
1982	-3,805.80	-2,230.87	-2,230.87	-3,812.09
1983	2,016.10	3,098.82	2,893.61	1,991.64
1984	-169.80	1,594.72	1,494.72	182.81
1985	3,569.40	5,385.40	5,272.14	2,994.58
1986	5,502.90	6,841.80	6,592.39	5,138.37
1987	5,874.60	7,522.20	7,368.83	5,462.11
1988	1,043.80	2,479.12	2,385.12	902.80
1989	-299.70	2,212.46	2,102.46	-369.70
Totals				
1972–79	4,224.2	34,627.3	32,258.7	4,414.5
1972–89	22,801.3	80,650.3	75,330.3	20,841.0
1979–83	2,969.6	25,646.8	23,196.0	2,717.7

Source: Author's calculations.

In the period 1972–89, the amount of capital flight varied from \$20.8 billion by the Duwendag method to a high of \$80.7 billion by the Morgan Guaranty method. It is not surprising that the Duwendag measure is exceedingly low because of the variables used. Given the relative importance of capital in capital-scarce economies like Nigeria, the most relevant definition of capital flight, which uses the sources-and-uses approach, is more appropriate, because it implicitly assumes that any outflow is abnormal because of the scarcity value of capital in developing countries. Concentrating on the World Bank definition, which is the more appropriate, implies that during the period 1972–79 total capital flight from Nigeria was \$22.8 billion. This figure will be used later in this chapter for purposes of comparison with the increases in external debt.

According to the World Bank approach, the years 1979, 1982, 1984, and 1989 were years of capital inflows. As to whether capital flight is

Table 6. Nigeria: Cross-Border Deposits of Nonbanks by Residence of Depositor, 1981-89

	Stock of Deposits	Change in Deposits	Ratio of Deposits to External Debt	Ratio of Deposits to GNP
	<i>(In millions of U.S. dollars)</i>		<i>(In percent)</i>	
1981	1,540		0.128	0.016
1982	1,380	150	0.108	0.015
1983	1,380	0	0.075	0.016
1984	1,170	-210	0.063	0.013
1985	1,500	230	0.078	0.017
1986	1,680	180	0.073	0.037
1987	2,300	620	0.077	0.096
1988	1,960	-350	0.063	0.068
1989	2,790	840	0.085	0.101
Total 1981-89	15,700	1,460		

Sources: IMF (1989, 1990); World Bank (1990).

episodic, it can be seen that during the period 1972-78, cumulative capital flight was \$4.3 billion. This was a period of military rule and one that coincided essentially with the peak in Nigeria's oil wealth syndrome. The total amount of capital flight during the civilian regime (1979-83) was \$3.0 billion, that is, 69 percent of that under the military regime. It is difficult, however, to come to any conclusion as to whether more capital flight actually occurred under the military regime than under the civilian regime, because the economic fortunes of Nigeria were not the same in the two periods.

The third approach is what was referred to earlier as the mirror stock statistics method. This method draws on international banking statistics to evaluate the amount of assets held abroad by residents of developing countries. This method of estimating capital flight has been used by Khan and Ul Haque (1987). It is particularly useful, as we shall see, in determining the minimum level of assets held abroad. For this method, the statistics recorded by the IMF are the cross-border bank deposits of nonbanks by residence of depositors, which for Nigeria are shown in Table 6. The statistics for Nigeria start in 1981. When capital flight is defined as the increase in this stock over the previous year, we find that the amount is relatively small. In all cases it is the lowest of all the estimates.

There are a number of reasons why this cannot be an adequate measure of capital flight. First, some funds are held in deposits outside the

major financial centers. Indeed, the nationality of depositors in some foreign banks is never revealed. The example most often cited is that of Swiss bank accounts, where secret codes are maintained to conceal not only the identity but also the nationality of the depositor. Second, substantial amounts that are not revealed are held in financial assets other than bank deposits, such as equities, bonds, or treasury bills, and in physical assets. As a result, the figures underestimate capital flight.

In a larger sense, however, these amounts are indicative of money that could have been used domestically. Such deposits are better seen within the context of other macroeconomic variables such as external debt and GNP (Table 6), where we find that the ratio of cross-border deposits to external debt varied from 6.3 percent to 12.8 percent. The ratio of cross-border deposits to GNP varied from 1.3 percent to about 10 percent for the period shown.

Causes of Capital Flight

The causes of capital flight as discussed in the literature are many. The various economic factors can be grouped under relative risks, exchange rate misalignment, financial sector constraints, fiscal deficits, and external incentives (Khan, 1989) and disbursements of new loans to developing countries (Cuddington, 1987). There are also other, noneconomic factors that, although important, are often ignored. These include corruption of political leaders and extraordinary access to government funds. These factors are now discussed.

In deciding where to invest, wealth holders look at the various risks of each prospective investment. There are certain inherent characteristics of developing countries that make risk attached to investments there larger than those of developed countries. Using the concept of expropriation risk within the context of an intertemporal optimizing model, Khan and Ul Haque (1985) show that any increase in risk in a rational expectations setting would tend to increase the outflow of private capital from the domestic economy into foreign countries where investments are less risky. This expropriation risk could include a variety of distortions, such as differences in taxes and political instability resulting in possible destruction of private property. Eaton (1987) builds on the Khan-Ul Haque model by relating the risk of expropriation of capital owned domestically, which is defined (especially in this case) as higher taxation, to public and publicly guaranteed foreign debt. The tax obligation arising from an increase in external debt can lead to capital flight. The flight of one investor leads to a rise in the potential tax obligations of the remaining investors. This also may create an incentive for other investors to move their assets abroad.

It is generally agreed that one of the principal causes of capital flight is exchange rate misalignment. It has been amply demonstrated in the empirical analysis of several studies (Dornbusch, 1985; Cuddington, 1986; Lessard and Williamson, 1987; Pastor, 1989, 1990) that the real exchange rate plays a significant role in the direction and magnitude of capital flight from highly indebted countries. Under normal circumstances, if a currency depreciation is expected, domestic wealth owners will shift out of domestic assets into foreign assets. In general, it is difficult to measure exchange rate expectations precisely. It is safe to assume, however, that if a currency is overvalued, economic agents will expect the currency to be devalued in the future. Holding firm to this expectation would cause residents to avoid the potential capital loss by converting domestic into foreign claims.

Financial sector constraints can lead to capital flight. It is well known that narrowness of capital and money markets is a feature of developing economies. These markets therefore provide only a limited variety of financial instruments in which wealth can be held. Many developing countries also lack full or credible deposit insurance on assets held in the domestic banking sector. As a result of these constraints, residents of developing countries look abroad to invest their wealth.

Additionally, there are extensive controls on interest rates and other aspects of financial market behavior in developing countries. Government policies in the financial sector have resulted in nominal interest rates that are far below the rates on comparable foreign financial instruments. In such situations it is expected that investors will seek alternative assets that will yield not only positive but higher returns.

Dornbusch (1985) has shown that capital flight is typically accompanied by a fiscal deficit. When a rising fiscal deficit is financed through the printing of money, this leads to inflationary pressure. To prevent the erosion of their monetary balances by inflation, investors move out of domestic assets as one way of avoiding the inflation tax. When a fiscal deficit is financed through bond sales, domestic residents may expect that at some future date their tax liabilities may increase to pay for the national debt. This would encourage domestic investors to move their assets to foreign countries to avoid potential tax liabilities.

Ize and Ortiz (1987) formalized the link between deficit financing and capital flight. In the Ize-Ortiz model, capital flight is related to the overall financial solvency of government. Insolvency and default risks created by a fiscal deficit appear explicitly as the determinants of capital flight.

A number of external factors influence the flight of capital, generally in terms of the opportunities available outside the country, including the

attractiveness of interest rates and the range of financial instruments in which wealth can be held. This is aptly put by Walter (1986, p. 20):

flight implies havens, and havens take the form of national status that provide an attractive range of real and financial assets to foreign based investors, political and economic stability, a favorable tax climate for nonresidents and various other attributes that generally are the obverse of conditions triggering capital flight in the first place.

Withholding taxes are not imposed on some types of nonresident deposits. Certain countries allow secret accounts, which are attractive to some wealth owners and can facilitate illegal transactions and tax evasion.

Under the principle of national sovereignty, it is difficult for foreigners to have inside information on asset holdings abroad. One safeguard is the domestic bank secrecy law, which bars both national and foreign authorities alike. The other is the blocking statute, which effectively prevents the disclosure, copying, inspection, or removal of documents located in the host country unless there is an order from or by foreign authorities.

Some authors argue that capital inflows in the form of disbursements to developing countries are a major cause of capital flight. In the case of public sector borrowing, the availability of foreign exchange increases the potential for graft and corruption. It is, therefore, logical to assert that for many developing countries (Nigeria included), abuse of official power can lead to capital flight. There is anecdotal evidence that highly placed public officials using the perquisites of their office siphon some of the money under their care to foreign countries solely for their private use.

In Nigeria's case it is difficult to rank the various causes of capital flight in any order of importance. However, a poor macroeconomic policy stance there has resulted in all kinds of distortions. At the same time, the role played by other factors such as access to foreign exchange through various perquisites of office and consequent possible abuse cannot be underestimated.

The Mechanisms of Capital Flight

There are many ways in which capital flight can occur. The conduits are varied, and it is almost impossible to develop an exhaustive inventory of channels. This section discusses the most significant channels for Nigeria.

First, transfers can take place through cash or monetary instruments. These are usually in the form of either foreign or domestic currency,

traveler's checks, or other checks. In the early 1970s, stories abounded about Nigerian currency being carried out of the country to major financial centers such as London and New York to be exchanged legally for other currencies at current market rates. In spite of the present economic predicament, there are still some African countries where the naira is exchanged for other currencies in the course of trade.

Second, capital flight can take place through bank transfers from a local affiliate of a foreign institution to a designated recipient abroad. This is possible at the market rate where no constraints or restrictions are in place. Transfers can still be possible in the face of exchange controls, but possibly at a less favorable rate. The history of the development of banking institutions in Nigeria shows the existence of local affiliates of foreign banks. That transfers of the type mentioned have been taking place in Nigeria cannot be in doubt. It is reasonable to claim, however, that such transfers may not be available for incomes that are illegally generated.

Another method of transfer is through precious metals and collectibles, including works of art. Local currency is converted into gold and silver or other precious metals, precious stones, jewelry, and similar assets that not only can be held abroad but will also retain their value. The sale value of these items is usually high in foreign currency, and governments generally tend to restrict or prohibit their import and export. Such international transfers therefore usually involve smuggling, with its inherent risks.

The fourth mechanism of transfer is through false invoicing of trade transactions, where export and import invoices issued differ from agreed prices. Estimates of this type of transfer have been undertaken by Bhagwati (1964), Naya and Morgan (1974), and Bhagwati, Krueger, and Welbulwasdi (1974). An analysis by Gulati (1987) shows that there can be systematic overinvoicing and underinvoicing of exports or imports. The expectation in the case of capital flight is that exporters will systematically engage in underinvoicing, whereas importers overinvoice, and in the process derive foreign exchange that is outside the control of the foreign exchange authority. The procedure is as follows: the foreign supplier issues an invoice that is greater than the agreed price of the product. The importer on receipt of the necessary foreign exchange remits it to the foreign supplier, who then keeps the difference in a bank for the use of the importer. On the export side, the invoice issued is for an amount in foreign currency that is less than the agreed price. The foreign buyer places the difference between the invoice price and the agreed price in a foreign bank account in the exporter's name and remits the invoice amount. It is this amount of money that is surrendered to the central bank for local currency at the

prevailing official exchange rate. To measure the magnitude of misinvoicing, partner country analysis is generally undertaken.

Capital flight through false trade invoicing generally involves the local affiliates of multinational companies and owners of businesses engaged in international trade. It is known in some cases that misinvoiced transactions are used to effect capital flight. This activity, in effect, arbitrages official and parallel-market exchange rates (Walter, 1986, p. 113).

A fifth method of transferring money abroad is through the black market, until recently a thriving source of such funds. The amount of money transferred in this way is difficult to estimate. Capital can also be transferred overseas through commissions and agents' fees, which are paid by foreign contractors into the foreign bank accounts of residents.

The Link Between Capital Flight and External Debt

Several authors have pointed out the link between capital flight and debt accumulation in developing countries. Indeed, some studies have shown that the ease with which capital flight takes place is related to the availability of foreign exchange. It is more appropriate, however, to examine the macroeconomic relationship between external debt and capital flight.

Within this context, the discussion can be along two lines. The first analyzes the relationship strictly in terms of causality between external debt and capital flight, whereas the second considers the various macroeconomic issues related to external indebtedness and capital flight. The literature indicates two kinds of linkages between external debt and capital flight. The first linkage runs from external debt to capital flight, and the second from capital flight to external debt. Each of these linkages can also be subdivided into two. Thus, the direct linkage can be divided into four major groups on the basis of whether the direction of causality runs from debt to capital flight or vice versa, and whether one simply provides the motive for the other or provides the means as well. The debt-to-flight mechanisms can be characterized as

- *Debt-driven capital flight.* If, consequent to external borrowing, residents of a country are motivated to move their assets to foreign countries, debt-driven capital flight is the result. Capital flees the country in response to attendant economic circumstances directly attributable to the external debt itself. The attendant economic circumstances leading to debt-driven capital flight are expectations of currency devaluation, fiscal crisis, pos-

sible crowding out of domestic capital, and avoidance of taxes and the risk of expropriation.

- *Debt-fueled capital flight.* In this case the inflow of capital provides both the motive and the resources for capital flight. In the case of debt-fueled capital flight, borrowed funds are themselves transferred abroad. There are two processes through which money can be transferred. First, government can borrow money and sell it to domestic residents, who then transfer the money abroad through legal or illegal means. In this case government is the provider of foreign exchange. Second, government can on-lend funds to private borrowers through a national bank, and the borrowers in turn transfer part or all the capital abroad. In this case the external borrowing provides the necessary fuel (the resources) for capital flight.

We now turn to an examination of causation in the other direction. There is, on the one hand, a case that is purely motivational, and on the other hand, a case where capital flight provides resources that reenter the country. These are referred to as flight-driven external borrowing and flight-fueled external borrowing, respectively:

- *Flight-driven external borrowing.* This situation develops when, as a result of capital that has left the country, there is a gap that needs to be filled in the domestic economy. Consequently, there is a demand for replacement on the part of both the government and the private sector. External creditors' willingness to meet this demand can be attributed to different risks and returns facing resident and nonresident capital.
- *Flight-fueled external borrowing.* In this situation domestic currency leaves the country but reenters in the guise of foreign currency. What happens is that the

flight capitalist seeks to arbitrage the yield and risk differentials between resident and external capital, by engaging in a series of transactions sometimes known as "round tripping" or "back to back loans." Resident capital is dollarized and deposited in an overseas bank, and the depositor then takes out a "loan" from the same bank (for which the deposit may serve as collateral) (Boyce, 1990, pp. 68–69).

The second set of arguments in this connection state that, when capital flees a country, the amount of money lost is equal to the potential investment in productive domestic activity. This investment would have earned foreign exchange if it had been made in the tradables sector of the economy. One popular argument calls for either an incentive to return funds held abroad by domestic residents or a significant re-

duction in the outflow of such funds. Accordingly, the heavily indebted countries would be in a better position for at least two reasons. The first is that the funds so returned can be used to boost domestic investment and thereby enhance debt-servicing capacity. Thus, the issue of capital flight is germane to the issue of real debt-servicing capacity. This is very important in the case of Nigeria because of its high debt-service ratio.

Second, a heavily indebted country that restricts capital flight would be in a better position to adjust to any subsequent fall in external funding. These two arguments are no doubt overstatements of the issues involved in capital flight. The impression is given that economic opportunities are equal between countries (which is not the case) and that the adoption of appropriate macroeconomic policies can release more funds domestically for investment purposes. Most of the arguments in the area are being increasingly linked to the issues of stabilization and growth (see, for example, Dornbusch, 1990). What is often ignored in the argument, however, is that it is possible for resources to be returned to the domestic economy in the form of financial holdings (fixed deposits, etc.), and not as machines and equipment for production or investment in the tradables sector.

There are other linkages between capital flight and external debt. One of the most popular hypotheses is the debt overhang argument, which states that large external debt discourages domestic investment. This is based on the fear that the tax liability of domestic investors will rise in the future.

Another relationship is that of internal transfer, where resources are transferred from the private to the public sector to finance government expenditure. As foreign debt increases, the internal transfer problem will also increase in magnitude, fostering capital flight.

A large external debt is a source of instability if for no other reason than that the outcome of the debtor-creditor position is hard to predict. As the fiscal burden of high external debt increases, a potentially unhealthy struggle for scarce resources within the economy is put in motion.

A better understanding of the relationship between capital flight and external debt can be gained by looking at some important statistics (Table 7). Using the capital flight figures derived earlier, we find the cumulative sum for the period 1972–89 is \$32.8 billion. The cumulative sum of the change in debt for the same period is \$32.2 billion. Thus, the ratio of capital flight to the change in external debt is about 79 percent for the entire period. For a year-by-year analysis, some selected periods have been chosen: in 1977, 1980, 1985, and 1986, capital outflows exceeded foreign debt accumulation, indicating the depletion of domestic resources.

Table 7. Nigeria: External Debt, Capital Flight, and Growth in GNP, 1972-89

	External Debt	Capital Flight	Change in External Debt	Ratio of Capital Flight to External Debt	Growth in GNP	Ratio of Gross Investment to GDP	Ratio of Capital Flight to Change in Debt
	<i>(In millions of U.S. dollars)</i>				<i>(In percent)</i>		
1972	732	106.4	81	14.5	15.0	18.3	131.4
1973	1,205	636.1	473	52.8	12.2	19.4	134.4
1974	1,274	325.0	69	25.5	74.1	14.6	471.0
1975	1,143	119.8	-131	10.5	20.8	21.8	-71.5
1976	906	124.8	-237	13.8	22.8	27.1	52.6
1977	3,146	2,490	2,240	79.1	13.7	26.7	111.2
1978	5,091	508.4	1,945	10.0	9.2	24.8	26.1
1979	6,235	-86.3	1,144	-1.4	25.5	20.1	-7.5
1980	8,934	2,713.3	2,699	30.4	29.0	20.5	100.5
1981	12,018	2,132.3	3,084	17.7	-5.6	21.4	69.2
1982	12,954	-3,805.8	936	29.4	-2.4	15.4	-406.6
1983	18,539	2,016.1	5,585	10.9	-3.5	11.4	36.1
1984	18,537	-169.8	-2.0	0.09	3.5	6.4	
1985	19,551	3,569.4	1,014	18.3	-3.7	7.5	352.0
1986	24,043	5,502.9	4,492	22.9	-48.2	11.4	122.5
1987	31,193	5,814.6	7,150	18.8	-49.2	12.3	81.4
1988	31,947	1,043.8	754	3.2	20.7	12.5	138.4
1989	32,832	-2,997.0	885	0.9	-1.2		
Total 1972-89		32,801.3	32,181				

Sources: International Monetary Fund, *International Financial Statistics Yearbook*, various issues; World Bank, *World Debt Tables*, various issues; and World Bank (1990).

It is significant that when the ratio of capital flight to the change in external debt was as high as 352 percent in 1985, the year preceding the adoption of the structural adjustment program in Nigeria, the investment-GDP ratio was 7.5 percent.

Important insights can be gained by looking at the changes in external debt and in investment and growth. In periods of rapid growth in external indebtedness, as in 1977–81, gross investment as a percentage of GNP was at the high end of its range of 20 to 27 percent for the entire period. Thus, increases in debt accumulation had an impact on investment. But the effect it could have had on growth in GDP was counteracted by capital flight. In 1981, 1983, 1985, and 1986, when the ratio of capital flight to the change in debt was 69 percent, 36 percent, 352 percent, and 122.5 percent, respectively, growth rates were negative 6 percent, 4 percent, 4 percent, and 48 percent.

These findings seem to lend credence to the general belief that capital flight has deterrent effects on the growth of the economy. The validity of such a statement and its definitiveness can only be based on rigorous empirical analysis beyond the scope of this study.

Conclusions

This chapter has addressed several general issues relating to capital flight. The magnitude of capital flight has been estimated using a number of alternative methods. The chapter has also discussed the causes and the mechanisms of capital flight and the link between capital flight and external debt, among other issues.

A number of conclusions can be drawn. The first is that there is no generally accepted definition of capital flight. Given the institutional framework and the nature of the economy, we have been able to adopt a definition and a measure of capital flight. Second, a significant proportion of total capital flight is recorded in the balance of payments and debt statistics. The implication of that, however, is that the adequacy of the measure depends on the accuracy of the items in the balance of payments statistics and the debt data. To the extent that these statistics are not accurate, their usefulness is in doubt.

Other important vehicles of capital flight are left out of the present estimates. Such vehicles include smuggling, currency movements, and misinvoicing. There are indications that a lot of money is transferred through trade misinvoicing. In general, since both the underinvoicing of exports and the overinvoicing of imports add to capital flight, the two should be added to estimate the net effect of trade misinvoicing on capital flight. To the extent that underinvoicing and overinvoicing exist in Nigeria, capital flight is underestimated. A third conclusion is

that if a large amount of domestically owned wealth resides abroad, tax revenues are adversely affected, and policy variables cease to be representative of the real situation.

Fourth, even though domestic policy distortions can lead to capital flight, the role of access to political office and the perquisites of office cannot be ignored. Indeed, many people who have transferred money abroad, and who belong to this category, do so not in the course of business but as a result of access to foreign exchange. The extent to which the provision of domestic incentives and the elimination of policy distortions will bring a reversal of capital flows is not precisely known. What is certain, however, is that political and macroeconomic stability plays an important role in the flow of capital. A suitable and stable macroeconomic environment that minimizes domestic macroeconomic policy errors will ensure that the economic distortions that bring about capital flight are eliminated.

Of significance in the area of policy errors that propel capital flight are inflation, exchange rate misalignment, fiscal deficit, and financial repression. The issue of corruption is more difficult for prescriptive purposes. The only safe thing that can be said is that there is a need for attitude changes, which require serious commitment to honest government on the part of political office holders.

Nigerians may own domiciliary accounts in which foreign currencies can be kept. The availability of this avenue may be important in the repatriation of some foreign funds. Its usefulness in attracting large sums of money to the country under the present system is limited, however.

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